

Volume 16

Web Admin, Club, School & Event Portal Architecture

v2 Technical Specification - Developer and Operator Ready

Purpose: define the web back-office and organizational portal layer for SkyThermal: platform administration, clubs, schools, event organizers, moderators, local experts, support teams and rescue operations. The portal converts the weather/AI/safety engine into daily operational workflows.

1. Document Control

Field	Value
Document	Volume 16 - Web Admin, Club, School & Event Portal Architecture
Version	v2 Technical
Primary audience	Product owner, CTO, front-end developers, back-end developers, DevOps, support operations, club/school managers, competition organizers
Dependencies	Volumes 01-15: PRD, SAD, database, API, weather engine, AI/ML, mobile UX, infrastructure, security, QA, business, legal, playbooks, hardware stations, in-flight safety
Delivery status	Implementation-ready architectural specification; not final UI pixel design or final legal text

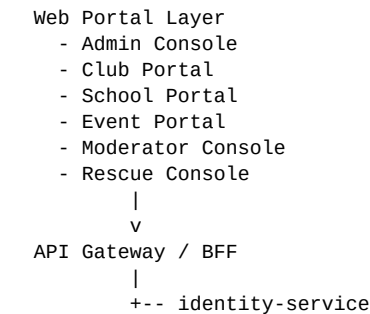
2. Executive Summary

The portal is the operating console for the paragliding intelligence platform. Unlike the mobile app, which focuses on pilot decisions, this web portal focuses on organizational operation: maintaining launch-site knowledge, moderating reports, managing memberships, running schools, planning competitions, watching safety dashboards, monitoring hardware weather stations and supporting paid organizations.

The portal should not be a generic admin panel. It should be a domain-specific command center for paragliding sites, weather operations and safety decisions. The UI must help a club safety officer answer: Which sites are flyable? Which reports are trusted? Which stations are down? Which students can fly? Which competition task is safest? Which pilots require follow-up?

Portal Area	Primary User	Main Outcome
Platform Admin	SkyThermal operations	Operate global platform, users, subscriptions, moderation, data quality and incidents
Club Portal	Club admin / safety officer	Manage local sites, members, reports, alerts, weather stations and playbooks
School Portal	School owner / instructor	Manage students, training windows, instructor decisions and student restrictions
Event Portal	Competition organizer / safety director	Plan tasks, weather briefings, live tracking, launch windows and safety communications
Moderator Console	Trusted local expert	Review reports, playbook edits, hazard submissions and false observations
Rescue Console	Authorized rescue/safety partner	View emergency share links, last known positions and search-area estimates

3. Architectural Position in the Platform



```

+-- organization-service
+-- entitlement-service
+-- launch-site-service
+-- playbook-service
+-- weather-service
+-- flyability-service
+-- event-service
+-- school-service
+-- moderation-service
+-- notification-service
+-- station-service
+-- tracking-service
+-- billing-service
+-- audit-service

```

The portal should use a Backend-for-Frontend (BFF) layer rather than directly calling every microservice. This keeps UI orchestration server-side, enforces tenant boundaries, simplifies entitlement decisions and allows screen-specific aggregation endpoints.

4. Design Principles

- Safety first: any operational decision screen must show confidence, age of data, uncertainty and blockers.
- Tenant isolation by default: club, school and event data must be scoped by `organization_id` and enforced server-side.
- Explainable operations: every moderation, site change, event weather decision and emergency access must have an audit trail.
- Map-first workflows: most portal functions should be spatially anchored to sites, regions, airspace, route tasks or station locations.
- Role-specific simplicity: a school instructor should not see global admin features; an event weather officer should not see billing unless allowed.
- Progressive enhancement: MVP works with simple tables and maps; V2 adds analytics dashboards, AI summaries and advanced operations.
- No hidden AI authority: AI can draft briefings and summarize risk, but human operators approve school, event and rescue actions.

5. User Roles and Permission Model

Role	Scope	Key Permissions	Sensitive Restrictions
super_admin	global	Manage all tenants, platform settings, source status, billing overrides, global moderation	Requires MFA, step-up auth for destructive actions
ops_admin	global/regional	View operational dashboards, manage data-quality incidents, station outages, support tickets	Cannot change billing or legal policy
support_agent	global/regional	View user/org support context, create support notes, resend invites	No location history without explicit support consent
club_owner	organization	Manage club members, sites, local alerts, station dashboards, playbook proposals	Cannot override global safety blockers
club_safety_officer	organization/site	Approve local reports, create safety advisories, manage site notes	All advisories audited and versioned
local_expert	site	Propose playbook edits, verify reports, flag hazards	Cannot publish changes without approval unless trusted threshold reached

school_owner	organization	Manage students, instructors, lesson calendar, subscription, school settings	Cannot access private pilot tracks unless shared
instructor	school/session	Create training sessions, assign student readiness, approve student fly/no-fly	Cannot alter subscription/billing
event_director	event	Create event, invite pilots, manage tasks, publish briefings	Task activation requires weather/safety confirmation
event_safety_officer	event	Monitor live tracking, issue safety notices, stop task	High-impact notices require confirmation
rescue_operator	incident	Access incident dashboard, last known positions, search-area export	Time-limited, reason-coded, audited access only

5.1 Permission Enforcement

```

authorization decision = allow if:
  token.valid == true
  AND user.status == active
  AND membership.org_id == requested_org_id
  AND membership.role has required_action
  AND entitlement permits portal_module
  AND resource.visibility allows role
  AND row-level policy allows organization/site/event scope
  AND high_risk_action passes step_up_auth when required

```

RBAC gives coarse access, ABAC adds resource-specific conditions such as site ownership, event scope, student assignment, emergency time window and legal consent. The front end may hide controls, but all permissions must be enforced by API Gateway/BFF and service layer.

6. Portal Information Architecture

Top Navigation	Primary Screens	MVP Priority
Overview	Operational dashboard, today conditions, alerts, station health, pending approvals	P0
Sites	Launch list, playbooks, hazards, landing zones, site reports, local experts	P0
Weather Ops	Model status, forecast validation, station observations, nowcast discrepancies	P1
Members	Org members, roles, invitations, access requests	P0
Reports	Pilot observations, flagged content, verification queue	P0
Schools	Students, instructors, training sessions, student readiness	P1
Events	Events, tasks, briefings, live tracking, scoring export	P1
Stations	Hardware station status, telemetry, calibration, maintenance tickets	P1
Billing	Plan, invoices, entitlements, payment portal	P0 for paid orgs
Settings	Organization profile, safety policy, notification routes, integrations	P0
Audit	Security logs, moderation logs, decision logs, export requests	P1

7. Web Front-End Architecture

- Recommended framework: Next.js or React/Vite for the first web portal; retain component logic compatible with the consumer map UI.
- Map rendering: MapLibre GL JS for sites, hazards, routes, tasks, station locations and airspace overlays.
- State model: server state with TanStack Query/SWR; local UI state with lightweight store; no business authorization logic only in client state.
- Design system: shared SkyThermal components from Volume 17, including risk badges, forecast tables, map layer controls, approval drawers and audit banners.
- BFF pattern: /portal/* endpoints return screen-specific aggregates to reduce client orchestration and authorization mistakes.
- Real-time updates: WebSocket/SSE for event live tracking, station health, urgent safety advisories and moderation queue changes.

```
/app
  /admin
    /overview
    /users
    /organizations
    /data-quality
    /support
  /club/[orgId]
    /overview
    /sites
    /reports
    /members
    /stations
  /school/[orgId]
    /students
    /sessions
    /instructors
  /event/[eventId]
    /dashboard
    /tasks
    /briefings
    /tracking
    /safety
  /rescue/[incidentId]
    /map
    /timeline
    /exports
  /components
    /Map
    /Forecast
    /Approval
    /Audit
    /Billing
    /Tables
  /lib
    apiClient.ts
    auth.ts
    entitlements.ts
    telemetry.ts
```

8. Portal BFF and Service Ownership

Capability	BFF Endpoint Family	Owning Service	Data Stores
Organization overview	/portal/orgs/{org_id}/overview	organization-service	organizations, memberships, entitlements
Site management	/portal/orgs/{org_id}/sites	launch-site-service	launch_sites, launch_playbooks, hazard_zones
Report moderation	/portal/moderation/*	moderation-service	pilot_reports,

			moderation_tickets, audit_events
Club members	/portal/orgs/{org_id}/members	organization-service	memberships, invitations
School sessions	/portal/schools/{org_id}/sessions	school-service	students, training_sessions
Event operations	/portal/events/{event_id}/*	event-service	events, tasks, briefings, tracking_summary
Station operations	/portal/stations/*	station-service	weather_stations, station_observations, station_health
Billing	/portal/billing/*	billing-service	subscriptions, invoices, entitlement_cache
Audit logs	/portal/audit/*	audit-service	audit_events

9. Platform Admin Console

9.1 Admin Overview Dashboard

Widget	Data Source	Operational Use
Global forecast ingestion status	model_runs, source_health	Detect late ECMWF/GFS/MET updates and downstream gaps
Safety alert queue	safety_advisories, alert_delivery	Confirm high-risk alerts are reviewed and delivered
Moderation backlog	moderation_tickets	Track pending playbook edits, hazard reports and content flags
Station network health	station_health_current	Identify offline/low-battery/miscalibrated stations
Subscription health	subscriptions, entitlements	Track paid org status, failed payments and plan mismatches
Support incidents	support_cases	Respond to org and pilot issues

9.2 Global Data Quality Operations

Weather data quality issues directly affect safety recommendations. The admin console must expose source freshness, forecast cube completeness, station QC failures, source-to-source disagreement and site-specific false-green reviews.

```
GET /portal/admin/data-quality/overview
Response:
{
  "source_health": [
    {"source":"ecmwf_open_data", "status":"healthy", "last_run":"2026-06-05T06:00:00Z", "latency_min": 38},
    {"source":"gfs_nomads", "status":"degraded", "last_run":"2026-06-05T00:00:00Z", "latency_min": 185}
  ],
  "forecast_cube_completeness_pct": 97.4,
  "false_green_reviews_open": 3,
  "station_qc_failures_24h": 18
}
```

10. Club Portal

10.1 Club Dashboard

The club portal focuses on local flying operations. Its primary page should show favorite club sites, today/tomorrow flyability windows, station health, pending local reports, active warnings, recent incidents and member activity.

Club Feature	Technical Requirement	Acceptance Criteria
Site watchlist	Org-site relationship table with display order and roles	Club admin can add/remove visible sites without global admin support
Local safety advisory	Advisory object with severity, site scope, expiry, author and approval status	Advisory appears on mobile app for affected pilots within 60 seconds
Report verification	Moderation queue scoped to club sites and trusted local experts	Verified report updates live reality layer and audit trail
Member roles	Membership table with role enum and permission grants	Club owner can invite safety officers and local experts
Station dashboard	Station health and observation widgets filtered by club-owned stations	Offline station creates maintenance ticket and alert

10.2 Club Site Playbook Workflow

1. Local expert proposes a playbook edit: wind direction range, rotor note, hazard zone, landing advisory or access rule.
2. System validates geometry, required fields, licence/provenance and duplicate conflicts.
3. Club safety officer reviews side-by-side diff against current published playbook.
4. If accepted, the playbook version changes from proposed to approved and then published.
5. Mobile clients receive invalidation event for affected site forecast and briefing cache.
6. AI briefing service uses the new playbook only after publication timestamp.

```
POST /portal/orgs/{org_id}/sites/{site_id}/playbook-proposals
{
  "change_type": "hazard_zone_add",
  "reason": "New rotor zone reported by three local pilots after tree line clearing",
  "geometry": {"type": "Polygon", "coordinates": [...]},
  "danger_conditions": {"wind_dir_from": 250, "wind_dir_to": 310, "min_wind_kmh": 18},
  "evidence_report_ids": ["rep_123", "rep_456"]
}
```

11. School Portal

11.1 School Operating Model

Schools need a workflow different from independent pilots. They must plan lessons, match conditions to student stage, document instructor decisions, keep student restrictions visible and avoid launching students under marginal conditions that might be acceptable for advanced pilots.

Object	Description	Important Fields
student_profile	A pilot under school supervision	skill_stage, wing_class, instructor_id, medical_notes_flag, active_restrictions
training_session	A scheduled lesson or flight block	site_id, start/end, weather_snapshot_id, instructor_id, status
student_readiness	Instructor assessment before launch	green/yellow/red, reasons, signed_by, timestamp
school_policy	School-specific safety limits	max_wind_kmh, max_gust_spread, cloudbase_min, allowed_sites

11.2 Student Decision Rules

```
student_allowed_to_fly = true only if:
  school_policy permits site
  AND training_session.status == active
  AND instructor assigned
  AND flyability.status in [GO]
  AND risk_score <= school_policy.max_student_risk
  AND wind_at_launch <= student_profile.max_wind_kmh
```

```

AND gust_spread <= school_policy.max_gust_spread
AND no hard safety blocker exists
AND instructor signs readiness check

```

The portal must record the weather snapshot used for each training decision. This protects the school, supports later debriefs and allows QA teams to investigate false-green or false-red cases.

12. Event and Competition Portal

12.1 Event Dashboard

Module	Purpose	Real-Time Requirement
Weather center	Forecast windows, model disagreement, cloudbase, thermals, storms, wind at task altitude	Refresh every forecast update; nowcast every 5-15 minutes where available
Task builder	Draw route, cylinders, turnpoints, start/goal, safety zones	Map edits saved with versioning and validation
Launch operations	Pilot check-in, launch open/closed, queues, wind limits	Live updates to pilots and safety team
Live tracking	Pilot positions, altitude, speed, last telemetry, emergency flags	WebSocket updates; degraded mode fallback
Safety desk	Weather alerts, airspace warnings, missing telemetry, SOS signals	High priority audible/visual alerts
Briefing publisher	AI-drafted, human-approved weather/task briefing	Versioned and delivered to event participants

12.2 Task Safety Validation

```

POST /portal/events/{event_id}/tasks/validate
{
  "task_geometry": {...},
  "start_window": "2026-06-05T11:00:00+02:00",
  "end_window": "2026-06-05T16:00:00+02:00",
  "pilot_class": "competition"
}

```

```

Response:
{
  "status": "MAYBE",
  "blockers": [],
  "warnings": [
    {"type": "model_disagreement", "segment": "turnpoint_2_to_3", "severity": "medium"},
    {"type": "valley_wind_after_15", "segment": "final_glide", "severity": "high"}
  ],
  "recommendation": "Move start 60 minutes earlier or shorten eastern leg."
}

```

13. Moderator Console

Queue Type	Inputs	Decision Options	Output
Pilot report verification	Observation, user reputation, nearby stations, forecast disagreement	verify, reject, request more evidence, escalate	verified observation or rejected report
Hazard submission	Geometry, notes, photos, reports, club evidence	approve draft, reject, merge duplicate, escalate	playbook proposal
Site edit proposal	Site metadata diff and provenance	approve, reject, request change	new site version
Abuse/spam report	Content, account history, device/IP indicators	dismiss, warn, suspend, ban	moderation action

The moderator console must present evidence context, not just raw text. For a wind report, show station observations, model forecast, nearby pilot tracks, report history and confidence. For a hazard polygon, show map overlay and affected launch directions.

14. Rescue and Safety Operations Console

Rescue mode is not a public admin feature. It is a highly restricted incident workspace. Access must be time-limited, reason-coded, approved where required and logged in immutable audit events.

Screen	Data Shown	Access Control
Incident overview	SOS state, pilot consent status, emergency contact route, last telemetry	incident-scoped rescue_operator only
Search map	Last known position, drift estimate, terrain, possible landings, route history	time-limited access token and audit reason
Timeline	Telemetry points, alerts sent, user responses, operator actions	read-only except operator notes
Export	PDF/GeoJSON/KML rescue packet	requires step-up auth and export reason

15. Billing and Entitlement Portal

Paid organizational plans require a portal surface for plan status, invoices, seats, entitlements and upgrades. Stripe Customer Portal can handle payment methods, invoices and subscription management, while SkyThermal owns internal entitlements, seat allocation, org roles and feature access.

```
POST /portal/billing/customer-portal-session
{
  "org_id": "org_123",
  "return_url": "https://app.skythermal.com/club/org_123/billing"
}

Response:
{"url":"https://billing.stripe.com/p/session/..."}
```

Plan	Portal Entitlements
Club	Club dashboard, members, local alerts, playbook proposals, station view, basic event tools
School	School dashboard, students, instructors, sessions, student-readiness decisions, training reports
Event	Event dashboard, task builder, live tracking, safety desk, briefing publisher
Rescue/Authority	Incident-scoped rescue console, export tools, limited live tracking access

16. Notification and Communication Workflows

- Club advisory: safety officer publishes site advisory; affected favorites receive push/email/in-app banner based on severity.
- School session change: instructor updates session go/no-go; students receive notification and calendar entry change.
- Event safety notice: safety officer sends task-wide alert; mobile app shows high-priority notice and WebSocket confirms delivery.
- Station outage: station-service triggers maintenance alert to club owners and ops team.
- Moderation escalation: risky report or hazard edit escalates to regional expert and platform ops.

```
notification_event = {
  "type": "event.safety_notice.published",
  "scope": {"event_id": "evt_456"},
  "severity": "high",
  "channels": ["push", "websocket", "email"],
  "requires_ack": true,
  "expires_at": "2026-06-05T16:00:00Z"
}
```

17. Database Extensions

```
CREATE TABLE organizations (
  id UUID PRIMARY KEY,
  org_type TEXT NOT NULL CHECK (org_type IN ('club','school','event_operator','rescue_partner','platform')),
  name TEXT NOT NULL,
  country_code CHAR(2),
  billing_customer_id TEXT,
  status TEXT NOT NULL DEFAULT 'active',
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

```
CREATE TABLE organization_memberships (
  id UUID PRIMARY KEY,
  org_id UUID NOT NULL REFERENCES organizations(id),
  user_id UUID NOT NULL REFERENCES users(id),
  role TEXT NOT NULL,
  scope JSONB NOT NULL DEFAULT '{}',
  status TEXT NOT NULL DEFAULT 'active',
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  UNIQUE (org_id, user_id, role)
);
```

```
CREATE TABLE portal_audit_events (
  id UUID PRIMARY KEY,
  actor_user_id UUID,
  org_id UUID,
  action TEXT NOT NULL,
  resource_type TEXT NOT NULL,
  resource_id UUID,
  before_state JSONB,
  after_state JSONB,
  reason TEXT,
  ip_hash TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

```
CREATE INDEX idx_memberships_org_role ON organization_memberships(org_id, role);
CREATE INDEX idx_portal_audit_org_time ON portal_audit_events(org_id, created_at DESC);
```

```
CREATE TABLE training_sessions (
  id UUID PRIMARY KEY,
  school_org_id UUID NOT NULL REFERENCES organizations(id),
  site_id UUID NOT NULL REFERENCES launch_sites(id),
  instructor_user_id UUID NOT NULL REFERENCES users(id),
  start_time TIMESTAMPTZ NOT NULL,
  end_time TIMESTAMPTZ NOT NULL,
  status TEXT NOT NULL DEFAULT 'planned',
  weather_snapshot JSONB,
  decision_status TEXT CHECK (decision_status IN ('GO','MAYBE','NO_GO')),
  decision_reasons JSONB,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

```
CREATE TABLE event_tasks (
  id UUID PRIMARY KEY,
  event_id UUID NOT NULL,
  name TEXT NOT NULL,
  task_geometry GEOMETRY(MULTILINESTRING, 4326),
  cylinders JSONB NOT NULL DEFAULT '[]',
  validation_status TEXT NOT NULL DEFAULT 'draft',
```

```

weather_snapshot JSONB,
published_at TIMESTAMPTZ,
created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX idx_event_tasks_geom ON event_tasks USING GIST(task_geometry);

```

18. API Contracts

Endpoint	Method	Purpose	Auth Scope
/portal/orgs/{org_id}/overview	GET	Organization dashboard aggregate	org:read
/portal/orgs/{org_id}/members	GET/POST	List/invite members	members:manage
/portal/orgs/{org_id}/sites	GET	Club/school visible sites	sites:read
/portal/orgs/{org_id}/sites/{site_id}/advisories	POST	Publish local advisory	advisory:publish
/portal/moderation/tickets	GET	Moderation queue	moderation:read
/portal/moderation/tickets/{id}/decision	POST	Apply moderation decision	moderation:decide
/portal/schools/{org_id}/sessions	GET/POST	Manage training sessions	school:sessions
/portal/events/{event_id}/tasks	GET/POST	Manage competition tasks	event:tasks
/portal/events/{event_id}/tracking	GET/WS	Live event tracking	event:tracking
/portal/rescue/incidents/{id}	GET	Incident workspace	rescue:incident

18.1 Standard BFF Response Envelope

```

{
  "data": {},
  "meta": {
    "request_id": "req_abc",
    "tenant_scope": {"org_id": "org_123"},
    "entitlements": ["club_dashboard", "site_playbooks"],
    "data_freshness": {"weather_age_min": 24, "station_age_sec": 42}
  },
  "warnings": [
    {"code": "MODEL_DISAGREEMENT_HIGH", "message": "Forecast confidence is reduced for this site."}
  ]
}

```

19. Real-Time Channels

wss://api.skythermal.com/ws/portal

```

subscribe messages:
{ "type": "subscribe", "channel": "org.org_123.station_health" }
{ "type": "subscribe", "channel": "event.evt_456.tracking" }
{ "type": "subscribe", "channel": "moderation.org_123.queue" }
{ "type": "subscribe", "channel": "incident.inc_789.timeline" }

event messages:
{ "type": "station.health.changed", "station_id": "st_1", "status": "offline" }
{ "type": "event.pilot.telemetry", "pilot_id": "p_1", "lat": 45.1, "lon": 11.2, "alt_m": 1450 }
{ "type": "moderation.ticket.created", "ticket_id": "mod_1", "severity": "medium" }

```

20. Security and Privacy Requirements

Threat	Required Control
BOLA across organizations	Server-side org_id enforcement, RLS where practical, contract tests for cross-tenant access
Unauthorized live tracking access	Event/incident scoped authorization, explicit participant consent,

	limited retention
Moderator abuse	Immutable audit events, dual approval for high-impact changes, role review
Billing privilege escalation	Billing actions limited to owner/admin roles and Stripe portal sessions tied to org
Emergency data exposure	Time-limited rescue access, reason codes, step-up auth, export logging
AI-generated unsafe briefings	Human approval for event/school briefings, data-grounding, no hidden automatic authority

21. Analytics and Operational Metrics

Metric	Why It Matters	Owner
club_weekly_active_admins	Measures adoption of club portal	Product
school_session_decision_coverage	Percentage of school sessions with stored go/no-go decision	Safety/Product
event_task_validation_usage	How often task validation is used before publication	Events
moderation_sla_minutes	Backlog and trust layer health	Operations
station_outage_time_minutes	Hardware network reliability	Operations
advisory_delivery_success_pct	Safety alert reliability	Platform
cross_tenant_access_blocked_count	Security monitoring	Security

22. UI Screen Specifications

Screen	Core Components	Critical States
Admin Overview	Source health cards, moderation queue, station health, incident panel, subscription status	degraded source, critical alert, backlog overflow
Club Overview	Site cards, local advisories, station widgets, pending reports, member activity	site NO_GO, station offline, report conflict
Site Playbook Editor	Map, hazard geometry editor, wind rule form, diff view, approval timeline	geometry invalid, duplicate hazard, requires approval
School Session Board	Session list, forecast snapshot, students, readiness buttons, instructor notes	student blocked, weather changed, instructor missing
Event Weather Center	Task map, forecast timeline, model disagreement, storm/lightning panel	task MAYBE/NO_GO, storm risk increasing
Event Tracking	Pilot map, telemetry table, safety alerts, missing pilot queue	SOS, stale telemetry, airspace violation
Rescue Incident Map	Last known position, drift cone, terrain, possible landings, export tools	export requires reason, token expired

23. Integration Points

- Identity provider: OIDC compatible provider such as Keycloak/Auth0/WorkOS; internal authorization still enforced by platform service.
- Billing: Stripe Billing/Customer Portal for subscription management, invoices and payment method updates; internal entitlements cached separately.
- Observability: Grafana dashboards for system metrics; portal may show selected embedded/linked operational dashboards to admins only.
- Calendar: optional ICS export for school sessions and event schedules.
- Map: MapLibre GL JS and vector tile sources for launch sites, tasks, hazards, stations and airspace.

- Email/push/SMS providers: notification-service abstracts channels; portal only configures policies and templates.

24. QA Acceptance Criteria

- Cross-tenant API tests prove a club admin cannot view another club, school, event or rescue incident.
- Every role has a documented permission test matrix with positive and negative cases.
- Event task validation returns deterministic status for golden weather/task fixtures.
- School session decision stores immutable weather snapshot and instructor decision audit event.
- Moderation actions require reason and create portal_audit_events records.
- Station outage updates portal dashboard within 60 seconds of station-service status change.
- Rescue incident export requires step-up authentication and records operator reason.
- Billing portal session cannot be created for non-owner roles.
- WebSocket reconnect logic preserves event tracking subscriptions after network loss.

25. MVP, V1, V2, V3 Roadmap

Phase	Deliverables
MVP	Org login, club dashboard, members, basic site list, report moderation, billing link, audit logging
V1	Site playbook editor, station dashboard, school sessions, event task drafts, real-time notification center
V2	Full event tracking dashboard, AI-drafted briefings with human approval, advanced moderation evidence panels, analytics
V3	Rescue console, multi-region operations, marketplace/admin integrations, advanced club/school compliance exports

26. Engineering Backlog

Epic	First Tickets
Portal shell	Create Next.js portal app; implement auth guard; implement org selector; implement role-aware navigation
Organization service	Add organizations/memberships/invitations schema; build member invite APIs; add audit events
Club dashboard	Build site watchlist API; create advisory workflow; render club site cards
Moderation	Create ticket queue; implement decision endpoint; create evidence panel UI
School portal	Add student/session schema; create training session board; store weather snapshot decisions
Event portal	Create event task schema; implement task validation API; add map task editor
Billing	Map org subscriptions to entitlement cache; implement Stripe portal session API
Real-time	Add portal WebSocket gateway; station health subscription; event tracking subscription
Security	Implement BOLA test suite; step-up auth for rescue/export; audit log viewer

27. References and Implementation Notes

- Stripe Customer Portal can handle hosted subscription management, payment method changes and invoices; the application still owns org entitlements and internal authorization.
- Keycloak Authorization Services and OIDC/RBAC patterns are suitable references for identity and resource-server authorization design, even if a managed provider is selected later.
- Grafana dashboards are useful for internal operational observability; customer-facing portal dashboards should expose curated domain data rather than raw infrastructure metrics.
- All high-impact safety actions must align with Volume 12 legal/safety rules and Volume 15 rescue-mode privacy constraints.